

Module 05: Lab 05

Exercises

1) In the ROC curve above, what is the TPR and FPR associated with the threshold value of 0.3?

TPR: [1] 0.6

FPR: [1] 0.2142857

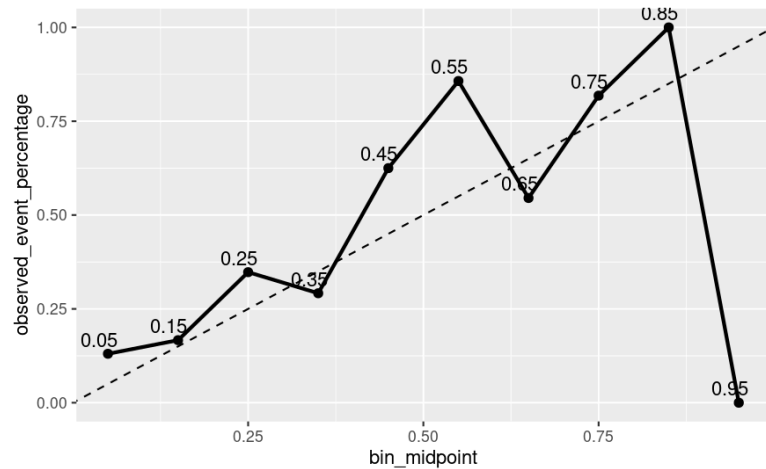
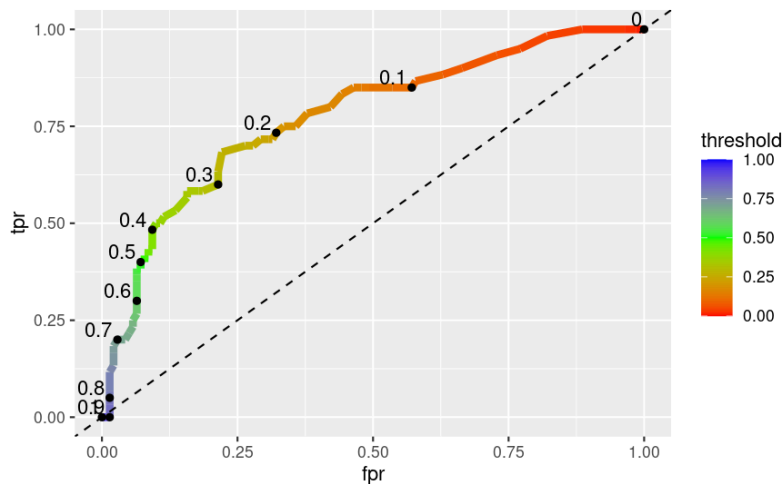
2) In the calibration curve above, are the predicted probabilities in the interval (0.2, 0.3) under-confident or over-confident?

```
167 {r}
168 interval_2 <-
  calibration_data[calibration_data$bin_midpoint == 0.25, ]
169 observed_event_percentage_2 <-
  interval_2$observed_event_percentage
170
171 print(observed_event_percentage_2)
172
173
```

[1] 0.3125

The probabilities are under-confident (greater than 0.25)

3) The 'AppliedPredictiveModeling' R package question.



```
# ROC curve
roc_data <- data.frame(threshold=seq(1,0,-0.01), fpr=0, tpr=0)
for (i in roc_data$threshold) {
  over_threshold <- lcp[lcp$Bad >= i, ]
  fpr <- sum(over_threshold$obs == "Good") / sum(lcp$obs == "Good")
  roc_data[roc_data$threshold == i, "fpr"] <- fpr
  tpr <- sum(over_threshold$obs == "Bad") / sum(lcp$obs == "Bad")
  roc_data[roc_data$threshold == i, "tpr"] <- tpr
}

ggplot() +
  geom_line(data = roc_data, aes(x = fpr, y = tpr, color = threshold), size =
2) +
  scale_color_gradientn(colors = rainbow(3)) +
  geom_abline(intercept = 0, slope = 1, lty = 2) +
  geom_point(data = roc_data[seq(1, 101, 10), ], aes(x = fpr, y = tpr)) +
  geom_text(data = roc_data[seq(1, 101, 10), ], aes(x = fpr, y = tpr, label =
threshold, hjust = 1.2, vjust = -0.2))

# Calibration curve
calibration_data <- data.frame(bin_midpoint=seq(0.05,0.95,0.1),
observed_event_percentage=0)
for (i in seq(0.05, 0.95, 0.1)) {
  in_interval <- lcp[lcp$Bad >= (i-0.05) & lcp$Bad <= (i+0.05), ]
  temp <- nrow(in_interval[in_interval$obs == "Bad", ]) / nrow(in_interval)
  calibration_data[calibration_data$bin_midpoint == i,
"observed_event_percentage"] <- temp
}

ggplot(data = calibration_data, aes(x = bin_midpoint, y =
observed_event_percentage)) +
  geom_line(size = 1) +
  geom_abline(intercept = 0, slope = 1, lty = 2) +
  geom_point(size = 2) +
  geom_text(aes(label = bin_midpoint), hjust = 0.75, vjust = -0.5)

...

```